SOMIN

ITMO UNIVERSITY

LECTURE 3 · PART 2 · ITMO UNIVERSITY · 28 OCTOBER 2026

Social Networks, Influence & Graphs

From six degrees and weak ties to PageRank, influence and graph neural networks.

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Lecturer · Tutors



RECAP · PART 1 AND YOUR HOMEWORK

Last week: one matching stack. Today: the graph under it.

PART 1 IN ONE PICTURE



Search, feeds and ads run the same funnel.

Two inputs we postponed: the graph — who follows, links to and buys with whom — and the random walks that rank it. That is today.

YOUR HOMEWORK · PEOPLE YOU MAY KNOW

How many mutual connections did your top ten suggestions have?

Most suggestions share friends with you: that is triadic closure, and it is how the algorithm finds them. LinkedIn's own experiment puts the sweet spot for jobs at about ten mutual connections. We get there in Section 2.

SOURCE Farseev, ITMO SMA Lecture 3, Part 1 (summary slide); homework from Part 1.

LECTURE 3, PART 2 · OUTLINE

Five questions for today.

01

Graphs

The vocabulary: paths, centrality, hubs, small worlds.

02

Ties

Closure, weak ties, brokers and homophily.

03

Influence & spreading

Who matters, how ideas spread, which seeds to pick.

04

Link analysis & communities

PageRank, link prediction, communities, fake networks.

05

Learning on graphs

Embeddings, GNNs, graph recommenders, GraphRAG.

Two old modules, rebuilt. The original ITMO graph theory lecture (after Easley & Kleinberg) and the influence and cascade parts of predictive analytics — with 2026 cases and graph machine learning.

SOURCE ITMO SMA modules 6 and 7; Easley & Kleinberg, Networks, Crowds, and Markets (2010), ch. 2–4, 14, 19–20.



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01

SECTION 1

Graphs: the language of networks

META · Q2 2026 RESULTS

3.6 billion people use Meta's apps every day. Every follow, like and message is an edge.

This section gives you the vocabulary: nodes and edges, paths and centrality, and the shapes real networks take — hubs and small worlds.

GRAPHS · NETWORKS ARE EVERYWHERE

Same maths for a karate club, the internet and Meta.

1977 · ZACHARY'S KARATE CLUB

34 members, 78 ties

A dispute split the club in two. Could the friendship graph predict the fault line? Section 4 answers.

1970 · ARPANET

13 nodes

The internet's ancestor: every link sits on a cycle, so one broken line never cuts anyone off.

2005 · HP LABS E-MAIL

436 employees

Who e-mails whom mirrors the org chart — and shows who really connects the teams.

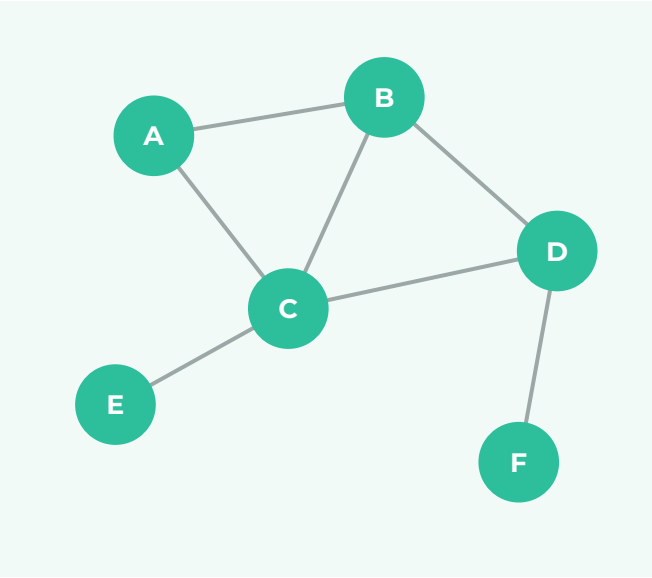
3.6B
PEOPLE A DAY
use Meta's apps (June 2026).

1.3B+
MEMBERS
on LinkedIn, the professional graph (Apr 2026).

41M
BLUESKY USERS
at the end of 2025 — an open graph anyone can read.

SOURCE Zachary, J. Anthropol. Research 1977; ITMO SMA Module 7 (ARPANET, HP e-mail after Adamic & Adar 2005); Meta Q2 2026; LinkedIn (Apr 2026); Bluesky transparency report (Jan 2026).

A graph is nodes and edges. So is your feed.



ADJACENCY MATRIX

	A	B	C	D	E	F
A	0	1	1	0	0	0
B	1	0	1	1	0	0
C	1	1	0	1	1	0
D	0	1	1	0	0	1
E	0	0	1	0	0	0
F	0	0	0	1	0	0

Real graphs store lists, not matrices: almost every cell is 0.

WEB

Pages → links

Directed. The graph PageRank runs on.

FOLLOW GRAPH

X, Instagram, TikTok

Directed: you can follow without being followed.

FRIENDSHIP

Facebook, LinkedIn

Undirected: both sides agree.

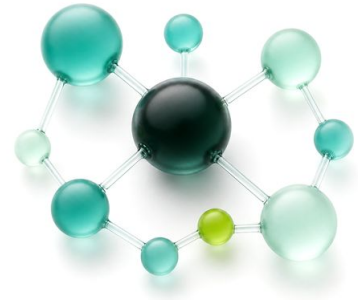
USER–ITEM

Users ↔ items

Bipartite: the recommender's graph (Part 1).

Meta: 3.6 billion daily people. As a matrix: 3.6B × 3.6B cells, nearly all empty.

SOURCE Easley & Kleinberg, Networks, Crowds, and Markets (Cornell, 2010), ch. 2; Farseev, ITMO SMA Module 7; Meta Q2 2026 results (3.60B daily active people).



NETWORKS · MEASURES

Four numbers that describe a node.

01

Degree

How many edges touch the node. In-degree (followers) and out-degree (following) differ.

02

Path length

Fewest hops between two nodes. Averaged over pairs, it tells how “small” the world is.

03

Clustering

Share of your friends who are friends with each other: $C = 2e / (k(k-1))$.

04

Centrality

Betweenness: shortest paths through you (bridges). Eigenvector / PageRank: linked from important nodes.

WORKED ON LAST SLIDE'S GRAPH

- ✓ **Node C:** degree 4 — the most connected.
- ✓ **Clustering of C:** 2 links (A–B, B–D) among its neighbours' 6 possible pairs: 1/3.
- ✓ **Path E → F:** 3 hops (E–C–D–F).

SOURCE Easley & Kleinberg (2010), ch. 2–3; Freeman 1977 (betweenness); Watts & Strogatz 1998 (clustering coefficient).

Real graphs carry direction, weight, sign and time.

DIRECTED

Who points to whom

Follows and hyperlinks go one way. In-degree (followers) and out-degree (following) tell different stories.

WEIGHTED

How strong the tie is

Messages per week, minutes watched, money sent. Weight is how we tell weak ties from strong ones.

SIGNED

Friend or foe

Likes and blocks, alliances and feuds. Balance theory: the enemy of my enemy tends to be my friend.

BIPARTITE

Two kinds of node

People and groups, users and items. Project it and you get a people-to-people graph of shared interests.

MULTIPLEX

Many relations at once

The same two people follow, message and buy together. Knowledge graphs mix dozens of relation types.

TEMPORAL

Edges with timestamps

Who knew whom, when. Cascades, rumours and job moves only make sense in time order.

SOURCE Easley & Kleinberg (2010), ch. 2, 4, 5; Stanford CS224W (Fall 2026), Lecture 1; ITMO SMA Module 7 (social-affiliation networks).

Getting the graph: formats, tools and access in 2026.

EDGE LIST	**A,B** **A,C** **B,C**
ADJACENCY LIST	**A: B, C** **B: A, C** **C: A, B**
SPARSE MATRIX	**only the 1s are stored (CSR format)**

1 billion edges stored as pairs of 64-bit IDs take about 16 GB before any features.

TOOLS

- **NetworkX** Python, perfect for teaching and graphs up to ~1M edges.
- **igraph** C core: much faster on large graphs.
- **PyTorch Geometric / DGL** graph neural networks (Section 5).
- **Neo4j** a graph database you query with Cypher.

ACCESS IN 2026

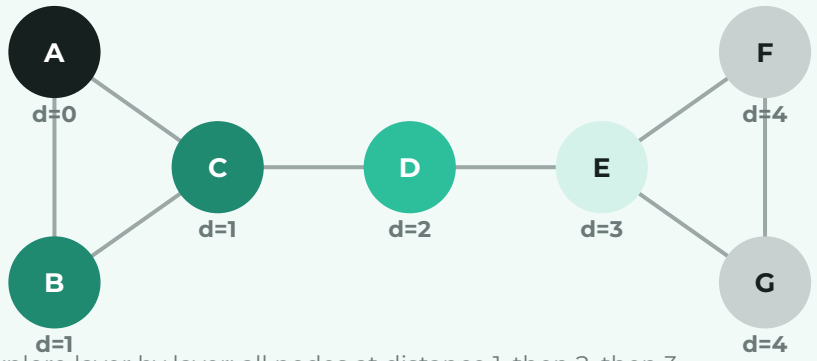
- **Bluesky:** the whole public firehose is open — no key, no fee.
- **X:** paid API; research access is contested.
- **EU DSA Art. 40:** vetted researchers can request platform data. Portal open since 29 Oct 2025; 49 applications under assessment by May 2026.
- **Meta, LinkedIn:** closed. Start from your own data export.

SOURCE AT Protocol docs, “Streaming data” (atproto.com); European Commission, DSA data-access roundtable (19 May 2026); Delegated Regulation (EU) 2025/2050; Lecture 1, slide 12.

GRAPHS · PATHS AND DISTANCE

Distance is the shortest path. BFS finds it.

BREADTH-FIRST SEARCH FROM A



Explore layer by layer: all nodes at distance 1, then 2, then 3...

Average distance between all pairs in this graph: 2.19 hops.

- **Path:** a sequence of nodes, each linked to the next.
- **Distance:** the length of the shortest path — A to F is 4.
- **Diameter:** the longest shortest path in the graph — 4 here.
- **BFS cost:** every node and edge once: O(V + E).

Shortest paths for all pairs are expensive on big graphs, so platforms sample.

MSN MESSENGER · 2008

6.6

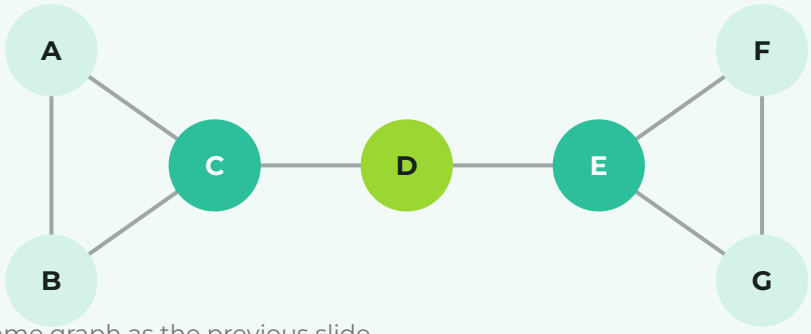
average hops between people

240 million people, 30 billion conversations in one month: a graph of 180M nodes and 1.3B edges. The most common distance was 6; 90% of pairs were within 7.8 hops.

SOURCE Leskovec & Horvitz, “Planetary-Scale Views on a Large Instant-Messaging Network”, WWW 2008; Easley & Kleinberg ch. 2; ITMO SMA Module 7. BFS example calculated by the lecturer.

GRAPHS · CENTRALITY, WORKED

Who is central? It depends on the question.



Same graph as the previous slide.

NODE	DEGREE	CLOSENESS	BETWEENNESS	EIGENVECTOR
A, B, F, G	2	0.40	0	0.33
C, E	3	0.55	8	0.45
D	2	0.60	9	0.38

Highest in each column in teal.

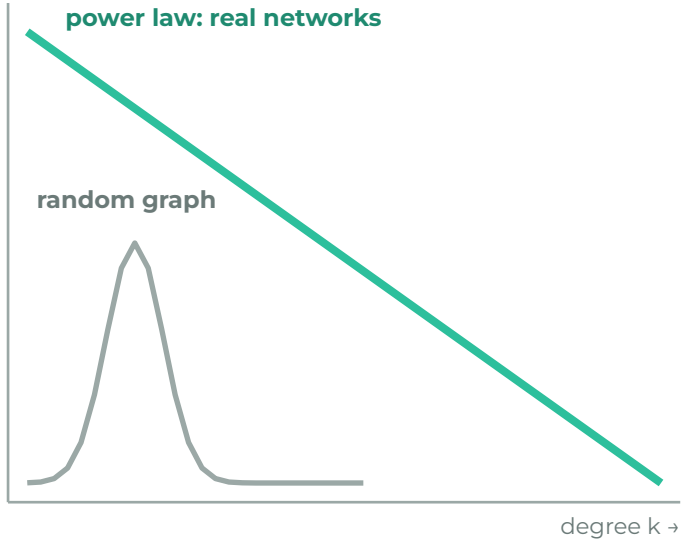
- **Degree:** how many friends.
- **Closeness:** 1 / average distance to everyone.
- **Betweenness:** shortest paths between other pairs that pass through you.
- **Eigenvector:** friends who are themselves central.

D has only two friends — and sits on 9 shortest paths. Remove D and the network splits in two. C and E have the most friends and the most central friends. Four measures, two different answers.

SOURCE Freeman 1977 (betweenness); Bonacich 1987 (eigenvector centrality); values calculated by the lecturer with NetworkX (unnormalised betweenness).

Hubs everywhere: the rich get richer.

HOW MANY NODES HAVE DEGREE k (LOG–LOG)



Barabási & Albert (1999): growth + preferential attachment — new nodes link to the already popular.

THE WEB'S BOW TIE · 203M PAGES, 1999



Core pages reach each other. IN pages reach the core but not back; OUT pages are reached but lead nowhere; 44M more hang off as tendrils. In-degree follows a power law with exponent ≈ 2.1.

WHY IT MATTERS HERE

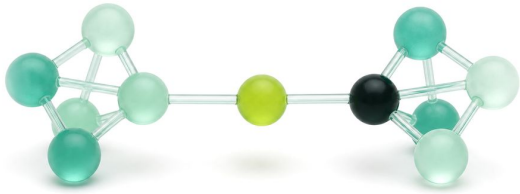
Popularity bias is preferential attachment you built yourself.

Shown → clicked → shown more. A recommender that only follows clicks makes hubs bigger and the long tail invisible.

SOURCE Barabási & Albert, Science 1999; Broder et al., “Graph structure in the Web”, Computer Networks 2000 (AltaVista crawl, May 1999: 203M pages, 1.47B links).

NETWORKS · SMALL WORLDS

Six degrees became 3.57.



1967–69

Milgram's letters

296 people in Nebraska and Boston tried to reach a stranger near Boston. 64 letters arrived, through 5.2 intermediaries on average.

1998

Watts & Strogatz

Highly clustered like a lattice, yet short paths like a random graph: a few random shortcuts are enough.

2011

Facebook, 721M users

Average distance 4.74 — that is, 3.74 people in between. “Four degrees of separation.”

2016

Facebook, 1.59B users

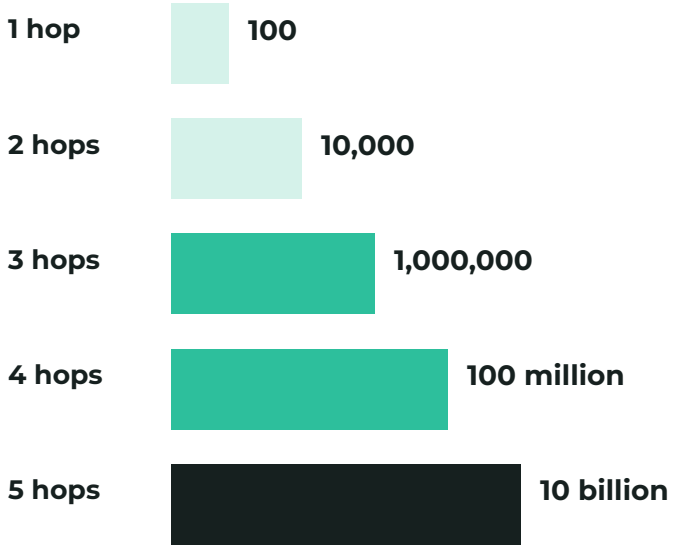
Average separation 3.57. The world got bigger and smaller at once.

Short paths + high clustering = news, rumours and memes cross the whole network in a few hops. It is also why HNSW (Part 1) finds the nearest vector in a few jumps.

SOURCE Travers & Milgram, Sociometry 1969; Watts & Strogatz, Nature 1998; Backstrom et al., “Four Degrees of Separation” (2012); Facebook Research blog (4 Feb 2016).

Branching makes paths short. Clustering fights back.

IF EVERYONE HAS 100 FRIENDS, NO OVERLAP



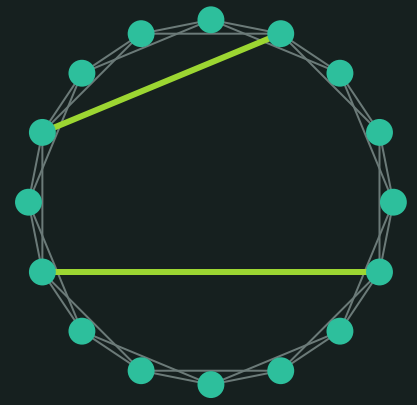
Five hops would exceed the world's 8.25 billion people.

BUT FRIENDS OVERLAP

Clustering slows the explosion

Your friends know each other (triadic closure, next section), so many friends-of-friends are people you already know. Pure clustering would make the world large again.

WATTS & STROGATZ · 1998



A clustered ring plus two random shortcuts (lime): path length collapses, clustering barely moves. A few long-range ties are enough.

SOURCE Easley & Kleinberg ch. 20; ITMO SMA Module 7 (“100 acquaintances” argument); Watts & Strogatz, Nature 1998; world population per DataReportal Digital 2026 (Lecture 1).

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02

SECTION 2

Ties: closure, weak ties and homophily

GRANOVETTER · 1973

People more often heard about their next job from an acquaintance than from a close friend.

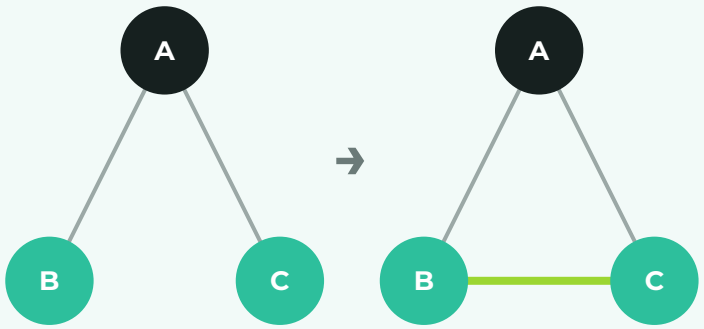
Edges are not equal. Strength, bridges and similarity decide what flows through a network: jobs, ideas, rumours — and recommendations.



TIES · TRIADIC CLOSURE

Friends of friends become friends.

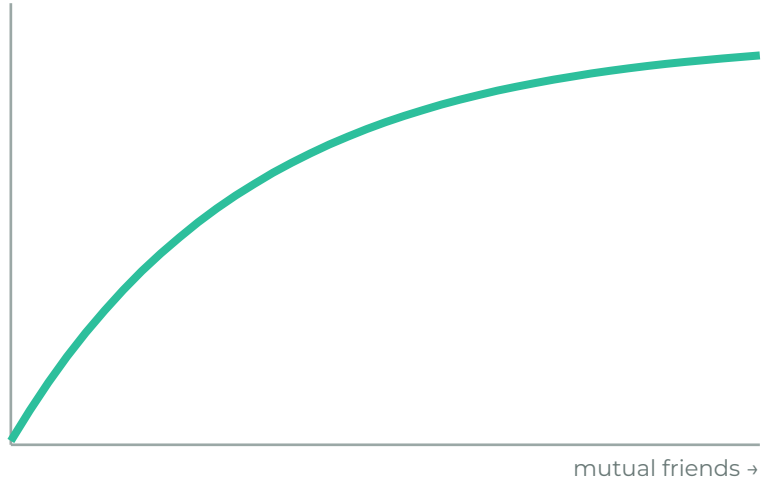
BEFORE → AFTER



STRONG TRIADIC CLOSURE

If A has strong ties to both B and C, then B and C have at least a weak tie.

CHANCE OF A NEW TIE VS MUTUAL FRIENDS (SHAPE)



Each extra mutual friend raises the chance that two people connect; shared classes work the same way (focal closure).

KOSSINETS & WATTS · SCIENCE 2006

43,553

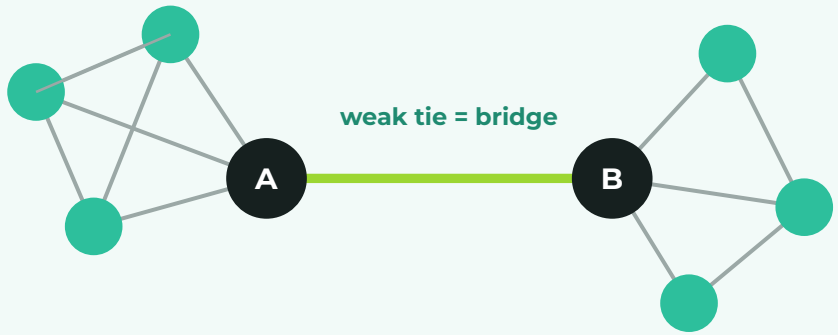
students, faculty and staff; e-mail over one academic year

The engine behind “people you may know”: rank strangers by how many friends you share — your homework numbers.

SOURCE Kossinets & Watts, “Empirical Analysis of an Evolving Social Network”, Science 311 (2006); Easley & Kleinberg ch. 3–4; ITMO SMA Module 7.

The strength of weak ties.

TWO CLUSTERS, ONE BRIDGE



Close friends know what you know. The acquaintance in another cluster knows what you don't.

GRANOVETTER, 1973

New information arrives through acquaintances

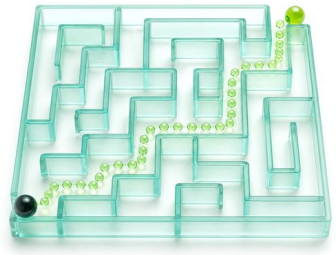
People more often heard about their next job from someone they saw rarely than from a close friend.

THE ARGUMENT, STEP BY STEP

1. **Strong ties close triangles.**
2. **So strong ties can't be bridges.**
3. **Bridges carry new information.**
4. **So weak ties carry it.**

Next: what exactly counts as a bridge — and a short proof of why it has to be a weak tie.

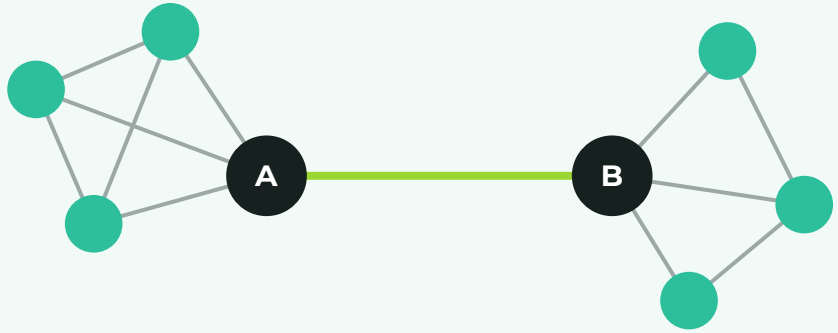
SOURCE Granovetter, “The Strength of Weak Ties”, AJS 1973; Easley & Kleinberg ch. 3; Adamic & Adar 2003; Liben-Nowell & Kleinberg, JASIST 2007 (link prediction).



TIES · LOCAL BRIDGES

Weak ties sit on bridges.

A–B IS A LOCAL BRIDGE: A AND B SHARE NO FRIENDS



- **Bridge:** removing the edge disconnects the graph.
- **Local bridge:** the two ends share no friends; its span is their distance once the edge is gone.
- **Neighbourhood overlap:** shared friends ÷ all friends of the two ends. Zero = local bridge.

WHY BRIDGES ARE WEAK

A short proof

Suppose A–B were strong and A had another strong friend C. Strong triadic closure forces a C–B tie — so A and B would share a friend and A–B would not be a local bridge. Hence: local bridges are weak ties.

PHONE CALLS · PNAS 2007

Remove weak ties and the network falls apart.

Call records of millions of mobile users: removing strong ties first barely hurt; removing weak ties first shattered the network.

SOURCE Easley & Kleinberg ch. 3; ITMO SMA Module 7 (local bridges, neighbourhood overlap); Onnela et al., “Structure and tie strengths in mobile communication networks”, PNAS 104 (2007).

NETWORKS · CASE: LINKEDIN, SCIENCE 2022

LinkedIn tested weak ties on 20 million people.

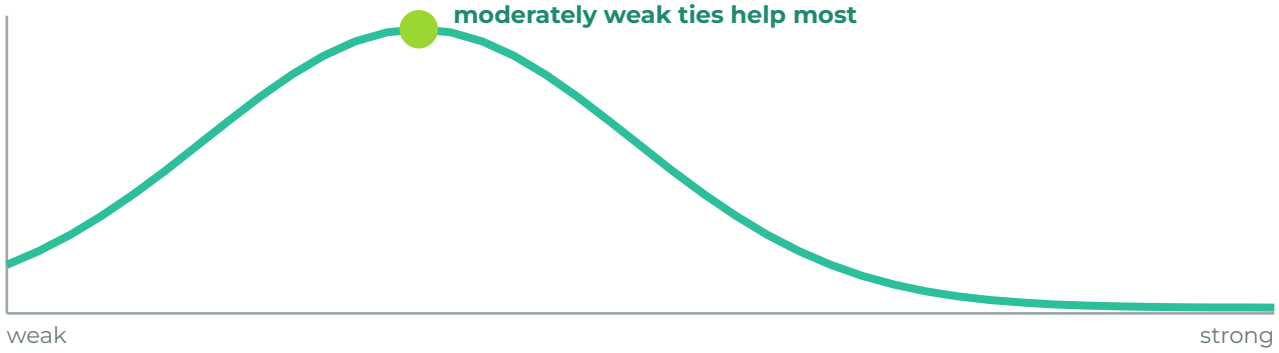
20M
USERS
in randomised experiments

5 yrs
OF DATA
of “people you may know”

2B
NEW TIES
created during the study

600K
NEW JOBS
traced back to the network

JOB MOBILITY VS TIE STRENGTH (SHAPE OF THE FINDING)



THE TWIST

The experiment was the recommender.

LinkedIn's “people you may know” varied the mix of weak and strong ties it suggested. So a ranking model changed who found jobs. Moderately weak ties — around 10 mutual connections — mattered most, and more so in digital industries.

SOURCE Rajkumar, Saint-Jacques, Bojinov, Brynjolfsson & Aral, “A causal test of the strength of weak ties”, Science 377 (2022); MIT Sloan (“magic number” of ~10 mutual connections).

TIES · HOW MANY FRIENDS DO YOU TALK TO?

Hundreds of friends, a handful of conversations.

~120

AVERAGE FRIENDS

per Facebook user in 2009 (median about 80).

4 / 6

TWO-WAY CONTACTS

in a month for the average man / woman.

10 / 16

WITH 500 FRIENDS

two-way contacts for men / women with 500 friends.

2–2.5×

MAINTAINED TIES

passively followed friends, versus the active network.

WHAT SOCIAL MEDIA REALLY GROWS

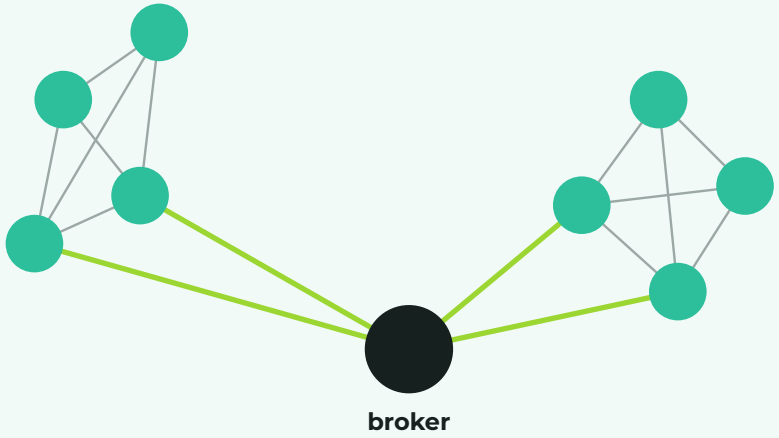
Not close friendships — those stay at a handful. It grows maintained ties: people you follow passively and rarely talk to. Those are the weak ties that carry new information, and they are exactly what a feed ranks for you.

SOURCE Marlow et al., Facebook Data Team (2009), reported in The Economist, “Primates on Facebook” (Feb 2009); Marlow, “Maintained relationships on Facebook” (Mar 2009); ITMO SMA Module 7.

TIES · STRUCTURAL HOLES

Brokers between groups have better ideas.

A BROKER SPANS A STRUCTURAL HOLE



No ties between the two groups — the hole. The broker sees both sides.

BURT · AJS 2004

673

managers in the supply chain of a large US electronics company

- more likely to express ideas
- ideas less often dismissed, judged more valuable
- higher pay, better reviews, more promotions

BETWEENNESS AT WORK

Managers whose networks spanned holes did better on every measure. Weak ties bring information; a bridging position turns it into ideas. Recall node D from Section 1.

SOURCE Burt, “Structural Holes and Good Ideas”, American Journal of Sociology 110(2):349–399 (2004).

Birds of a feather — or contagion? Hard to tell apart.

HOMOPHILY · 2001

Similarity breeds connection

We link to people like us: race and ethnicity divide most, then age, religion, education, occupation and gender.

CONTAGION · NEJM 2007

Behaviour seems to spread

12,067 people followed for 32 years: your chance of becoming obese rose 57% if a friend became obese.

THE CATCH · 2011

The two are confounded

In observational network data, homophily and contagion are “generically confounded”. Only experiments — like LinkedIn's — separate them.

WHAT IT MEANS FOR YOUR WORK

- **For analytics:** friends who behave alike are not proof of influence.
- **For recommenders:** social filtering works partly because friends were alike to begin with.

SOURCE McPherson, Smith-Lovin & Cook, Annual Review of Sociology 2001; Christakis & Fowler, NEJM 2007; Shalizi & Thomas, Sociological Methods & Research 2011.

Selection or influence? Three kinds of closure.

01 · TRIADIC CLOSURE

A friend in common

Two people with a mutual friend tend to connect. The more mutual friends, the likelier.

02 · FOCAL CLOSURE

A focus in common

A shared class, club or employer. In the university data, one shared class worked like shared friends — with diminishing returns.

03 · MEMBERSHIP CLOSURE

Joining your friends' group

LiveJournal, 2006: the chance of joining a community rises with friends already in it — and more if those friends know each other.

Why it matters: selection (similar people connect) and influence (connected people become similar) call for different action. Teen smoking, from the original course: if it is influence, target the key players; if it is selection, target the behaviour itself.

SOURCE Kossinets & Watts, Science 2006 (focal closure); Backstrom, Huttenlocher, Kleinberg & Lan, KDD 2006 (LiveJournal); Easley & Kleinberg ch. 4; ITMO SMA Module 7.

TIES · CASE: TWITTERRANK'S SINGAPORE DATA

Follow-back is homophily you can measure.

72.4%

FOLLOW BACK

of users follow more than 80% of their followers.

80.5%

ARE FOLLOWED BACK

by 80% of the people they follow.

6,748

USERS

around 996 top Singapore accounts; 1,021,039 tweets.

WHY DO PEOPLE FOLLOW BACK?

Users who follow each other tweet about more similar topics than users who don't: reciprocity is explained by homophily. And it inflates follower counts — one more reason followers aren't influence (Section 3).

ASSORTATIVITY · NEWMAN 2002

Do hubs link to hubs?

One number from −1 to +1. Social networks are usually assortative: well-connected people know other well-connected people. Technological and biological networks are disassortative: hubs connect to many small nodes.

SOURCE Weng, Lim, Jiang & He, “TwitterRank”, WSDM 2010; ITMO SMA Module 6 (dataset); Newman, “Assortative mixing in networks”, Phys. Rev. Lett. 89 (2002).

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03

SECTION 3

Influence and spreading



CHA ET AL. · ICWSM 2010

“Popular users who have high indegree are not necessarily influential.”

Who matters for a topic, how messages spread, which seeds to pick for a campaign — and whether a cascade can be predicted from its first few shares.

NETWORKS · INFLUENCE

Followers are not influence.



THE MILLION FOLLOWER FALLACY · ICWSM 2010

55M
TWITTER USERS
1.96B follow links, 1.76B tweets

0.55 → 0.12
FOLLOWERS VS RETWEETS
rank correlation, all users → top 10%

2 of 20
SHARED TOP-20
followers, retweets and mentions agree on 2

Among the most-followed accounts, follower count barely predicts who gets retweeted.

TWITTERRANK · WSDM 2010

Influence is topic-specific: PageRank run per topic.

EVERYONE'S AN INFLUENCER · 2011

74M diffusion events: many ordinary users beat a few stars on cost.

FOR MARKETERS

Buy influence on a topic, not followers in general.

- ✓ **Measure what spreads:** reshares and mentions, not audience size.
- ✓ **Match the topic:** a fitness creator moves fitness, not fintech.
- ✓ **Spread the bet:** many mid-size creators are cheaper per cascade.

SOURCE Cha, Haddadi, Benevenuto & Gummadi, ICWSM 2010; Weng et al., WSDM 2010 (TwitterRank); Bakshy, Hofman, Mason & Watts, WSDM 2011.

INFLUENCE · TWITTERRANK

TwitterRank: PageRank, topic by topic.

01

Distil topics

Run a topic model (LDA) over each user's tweets: every user gets a topic mix.

02

Weight the follow graph

For topic t, the surfer moves from a follower to a friend in proportion to the friend's tweets and their topic similarity.

03

Walk per topic

A random walk with teleport, one per topic, gives TR□: each user's influence on topic t.

THE TOPIC-SPECIFIC RANK

$$TR_t = \gamma \cdot P_t \cdot TR_t + (1 - \gamma) \cdot E_t$$

P□: topic-weighted follow transitions. E□: teleport toward users who tweet a lot on t.

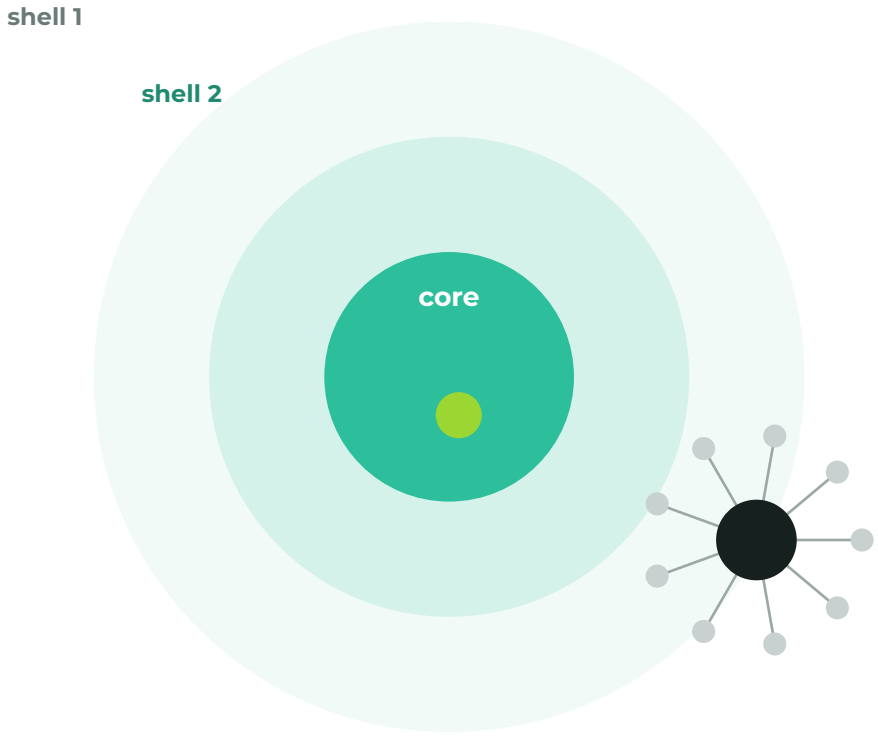
TESTED ON "WHOM TO FOLLOW"

Hide real follow links and ask each method to recover them. TwitterRank beat in-degree, PageRank and topic-sensitive PageRank in 5 of 8 scenarios on the Singapore data.

SOURCE Weng, Lim, Jiang & He, "TwitterRank: finding topic-sensitive influential twitterers", WSDM 2010; ITMO SMA Module 6.

INFLUENCE · THE CORE MATTERS

Spreaders sit in the core, not just at the hubs.



Dark: a hub on the edge. Lime: a modest node in the core.

K-SHELL DECOMPOSITION

- ✓ Remove every node with degree 1, repeatedly: that is shell 1.
- ✓ Then degree 2, and so on.
- ✓ Whatever survives longest is the core.

NATURE PHYSICS · 2010

In spreading simulations on real networks, the most efficient spreaders sat in the core — not necessarily the nodes with the highest degree or betweenness. A hub on the periphery reaches many people who don't pass it on.

SOURCE Kitsak et al., “Identification of influential spreaders in complex networks”, Nature Physics 6:888–893 (2010).



SPREADING · TWO MODELS

How a message spreads: cascade or threshold.

INDEPENDENT CASCADE

One chance per neighbour

When you become active, you get a single chance to activate each neighbour, succeeding with probability p. Good for information: one exposure can be enough.

LINEAR THRESHOLD

Enough of my friends did it

Each person has a threshold θ. They adopt once the weighted share of adopting neighbours passes θ. Good for behaviour: you need reinforcement.

CENTOLA · SCIENCE 2010

54% vs 38%

1,528 people in an online health community. Signing up for a health forum reached 54% in clustered networks versus 38% in networks of long ties, and spread about four times faster.

Simple vs complex contagion. A rumour can cross one weak tie. A costly behaviour needs several friends doing it first — so clustering helps it spread.

SOURCE Kempe, Kleinberg & Tardos, KDD 2003 (IC and LT models); Centola, “The Spread of Behavior in an Online Social Network Experiment”, Science 329 (2010).

SPREADING · INFLUENCE MAXIMISATION

Pick k seeds to start the biggest cascade.

01

The problem

Choose k people to activate first so the expected spread (IC or LT) is largest. Finding the best set is NP-hard.

02

Greedy

Add the person with the largest marginal gain in spread. Repeat k times.

03

The guarantee

Greedy lands within 63% (1 − 1/e) of the optimum, because spread is submodular: each extra seed adds less.

04

Make it fast

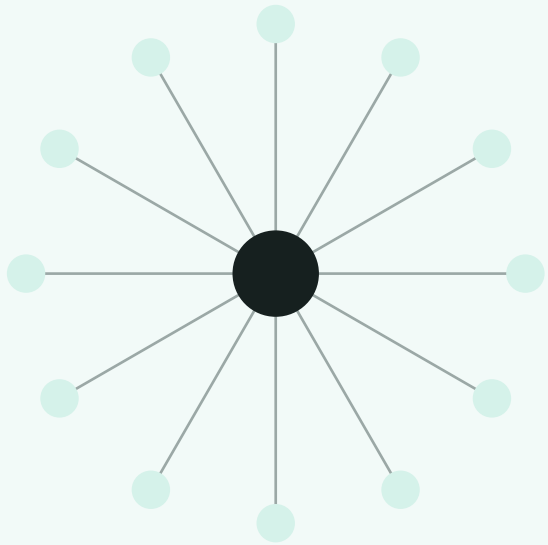
Lazy evaluation (CELF) skips seeds whose gain can't win: about 700× faster.

For campaigns: the second creator in the same community adds little — diminishing returns, made precise. Pick seeds across different communities, then weigh cost per seed (Section 1's centralities and Section 4's communities are your inputs).

SOURCE Kempe, Kleinberg & Tardos, “Maximizing the spread of influence through a social network”, KDD 2003; Leskovec et al., “Cost-effective outbreak detection in networks”, KDD 2007 (CELF).

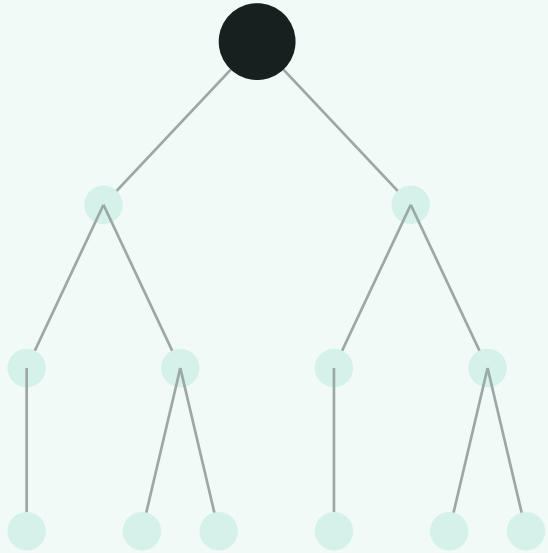
Most “viral” content is really one big broadcast.

BROADCAST



One account, many direct re-sharers. Depth 1.

VIRAL



Person to person, generation after generation.

1 BILLION DIFFUSION EVENTS

Popularity is driven by the size of the largest broadcast.

Structural virality on Twitter is typically low, whatever the size. Big cascades usually mean one big account shared it — so who amplifies matters more than the chain.

SOURCE Goel, Anderson, Hofman & Watts, “The Structural Virality of Online Diffusion”, Management Science 62 (2016); Farseev, ITMO SMA Module 6 (popularity prediction).

SPREADING · CAN CASCADES BE PREDICTED?

Watch the first five reshares, predict the rest.

WILL THE CASCADE REACH THE MEDIAN SIZE (≈ DOUBLE)? · ACCURACY



AUC 0.877 with all features. Temporal ≈ 0.73 is approximate (read from the paper's figures).

- ✓ **Speed is the signal:** fast early reshares predict big cascades.
- ✓ **Breadth beats depth:** early on, wide cascades grow more than long chains.
- ✓ **Content alone is weak.** Who shares and how fast matter more.

FACEBOOK PHOTOS · WWW 2014

150,572

photos, 9.2 million reshares, June 2013

Observe the first k reshares, then predict whether the cascade doubles. Prediction gets easier the more you observe: “see 20, predict 40” beats “see 5, predict 10”. Lab 2 offers virality prediction: these are your features.

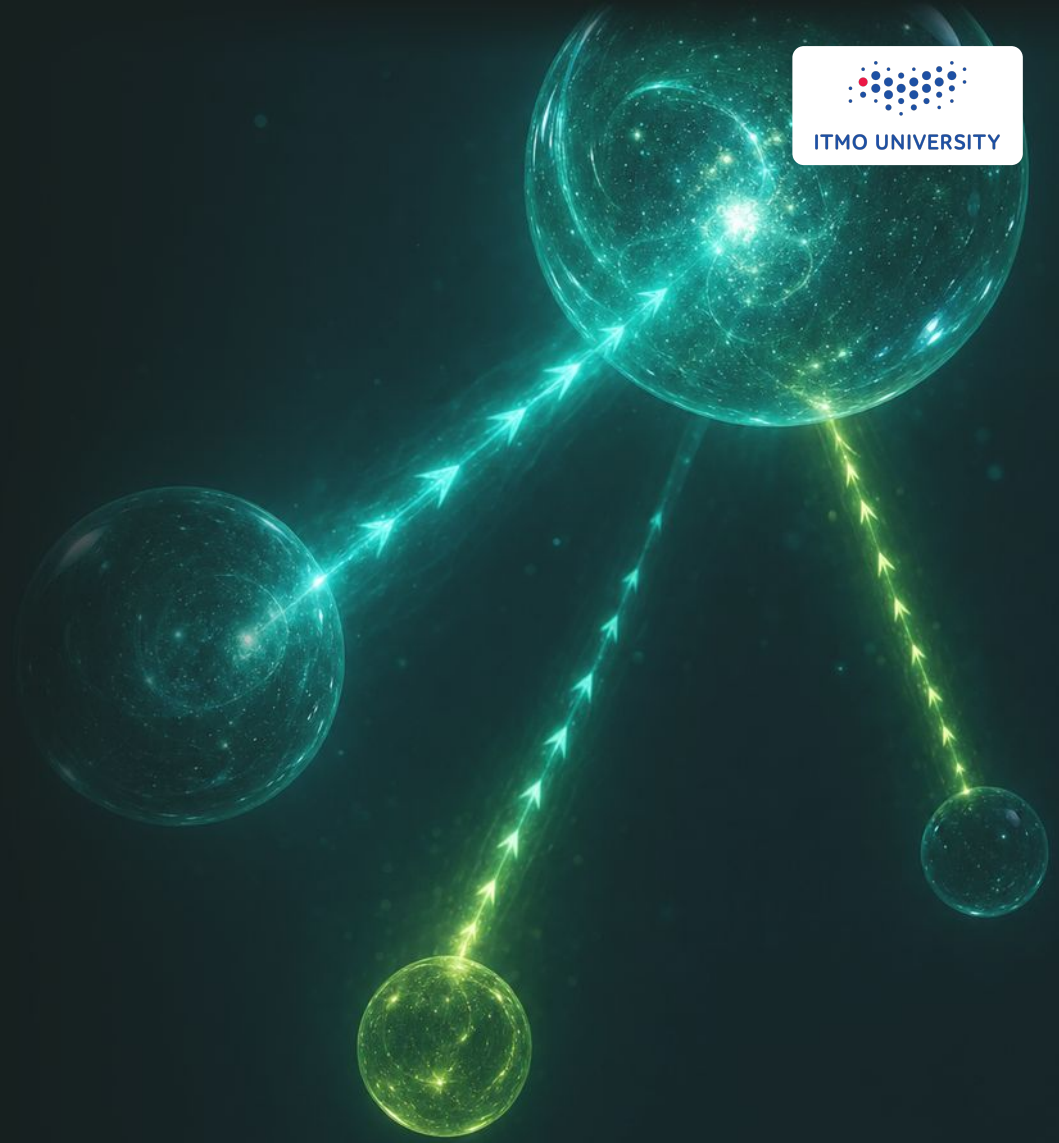
SOURCE Cheng, Adamic, Dow, Kleinberg & Leskovec, “Can cascades be predicted?”, WWW 2014; ITMO SMA Module 6.

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04

SECTION 4

Link analysis and communities



GOOGLE · DECEMBER 2025

“How PageRank works has evolved a lot since then, and it continues to be part of our core ranking systems.”

Ranking nodes by who links to them, predicting links that don't exist yet, finding communities — and catching the fake ones.

PageRank: important pages are linked from important pages.

BRIN & PAGE, WWW 1998 · d = 0.85

$$PR(p) = (1 - d) / N + d \cdot \Sigma\ PR(q) / out(q) \quad \text{over pages } q \text{ that link to } p$$

- **Random surfer.** Follow a random link with probability 0.85; jump to a random page with 0.15.
- **Score = time share.** PageRank is the fraction of time the surfer spends on each page in the long run.
- **Votes are weighted.** A link from an important page counts more; a page with many links splits its vote.
- **Power iteration.** Start everyone equal; redistribute; repeat until the numbers stop moving.

STILL IN USE

“How PageRank works has evolved a lot since then, and it continues to be part of our core ranking systems.”

Google, Guide to Search ranking systems (Dec 2025)

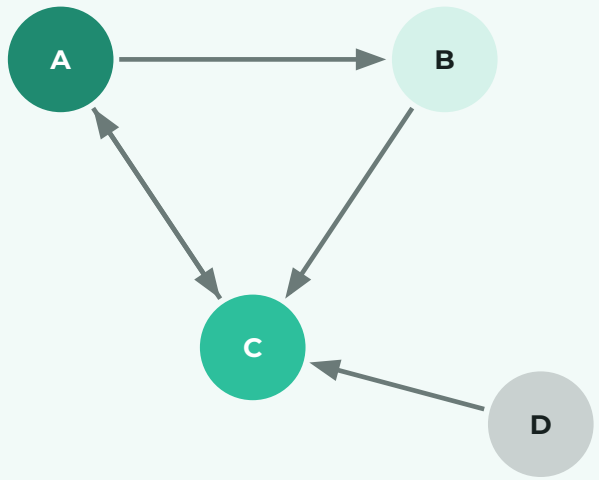
The same idea ranks people (TwitterRank), scientific papers (citation counts weighted by source) and items in a recommender graph.

SOURCE Brin & Page, “The Anatomy of a Large-Scale Hypertextual Web Search Engine”, WWW7 (1998); Google Search Central, ranking systems guide (updated 10 Dec 2025).

NETWORKS · PAGERANK, WORKED

Four pages, a few iterations, one ranking.

LINKS



A→B, A→C, B→C, C→A, D→C. N = 4, d = 0.85.

ITERATION	A	B	C	D
0	0.250	0.250	0.250	0.250
1	0.250	0.144	0.569	0.038
2	0.521	0.144	0.298	0.038
3	0.291	0.259	0.413	0.038
converged	0.373	0.196	0.394	0.038

Each row sums to 1. The numbers swing, then settle.

READ THE RESULT

C wins: three pages link to it.

A is a close second with one in-link — because it comes from C.

D gets only the teleport share: 0.15 / 4 = 0.0375.

SOURCE Power iteration calculated by the lecturer (60 iterations to convergence) with the formula on the previous slide.

NETWORKS · RANDOM WALKS RECOMMEND

Personalise the random walk and PageRank recommends.

HITS · KLEINBERG 1999

Hubs and authorities

A good hub links to many good authorities; a good authority is linked from many good hubs. Two scores that reinforce each other.

PERSONALISED PAGERANK

Always teleport home

Let the surfer jump back to you instead of to a random page. The scores now measure what is close to you in the graph: accounts, pages, products.

IN PRODUCTION

Who to follow, what to pin

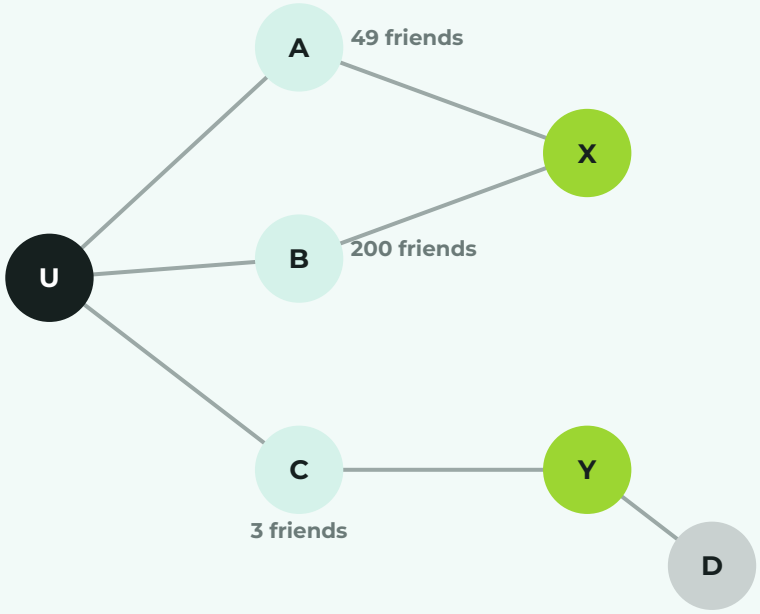
Twitter's "Who to Follow" (WWW 2013) ran random walks from each user's circle of trust. Pinterest's PinSage (2018) samples neighbours by random walks on a 3-billion-node graph.

One surfer, three jobs: link analysis ranks pages, link prediction suggests people, and walks on the user–item graph recommend items.

SOURCE Kleinberg, J. ACM 1999 (HITS); Gupta et al., "WTF: The Who to Follow Service at Twitter", WWW 2013; Ying et al., "PinSage", KDD 2018.

LINK ANALYSIS · PEOPLE YOU MAY KNOW, WORKED

Three ways to score a link that doesn't exist yet.



Should U meet X or Y?

CANDIDATE	COMMON	JACCARD	ADAMIC–ADAR
X (via A, B)	2	0.67	0.45
Y (via C)	1	0.25	0.91

Highest score per column in teal.

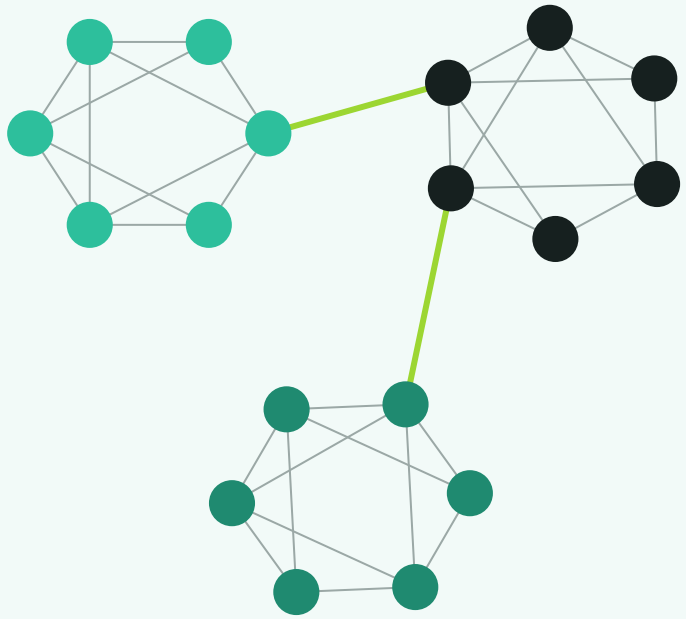
- **Common neighbours:** $|N(u) \cap N(v)|$
- **Jaccard:** $|N(u) \cap N(v)| \div |N(u) \cup N(v)|$
- **Adamic–Adar:** $\Sigma\, 1 / \log(\text{degree})$ over shared friends

Adamic–Adar flips the ranking. A shared friend who knows three people says more than two hubs who know everyone — idf again.

SOURCE Liben-Nowell & Kleinberg, JASIST 2007 (link prediction); Adamic & Adar, Social Networks 2003; values calculated by the lecturer.

NETWORKS · COMMUNITIES

Communities: dense inside, sparse between.



Lime edges: the bridges Girvan–Newman cuts first.

GIRVAN–NEWMAN · 2002

Cut the bridges

Remove the edge with the highest betweenness, recompute, repeat until clusters fall apart.

LOUVAIN · 2008

Greedy modularity

Merge nodes while the share of within-community edges rises. Ran on 118M nodes and 1B+ links.

LEIDEN · 2019

Louvain, fixed

Louvain left up to 25% of communities badly connected. Leiden guarantees connected ones, and is faster.

IN PRODUCTION · X

SimClusters

About 145,000 communities used to find out-of-network posts you might like.

WHY IT MATTERS

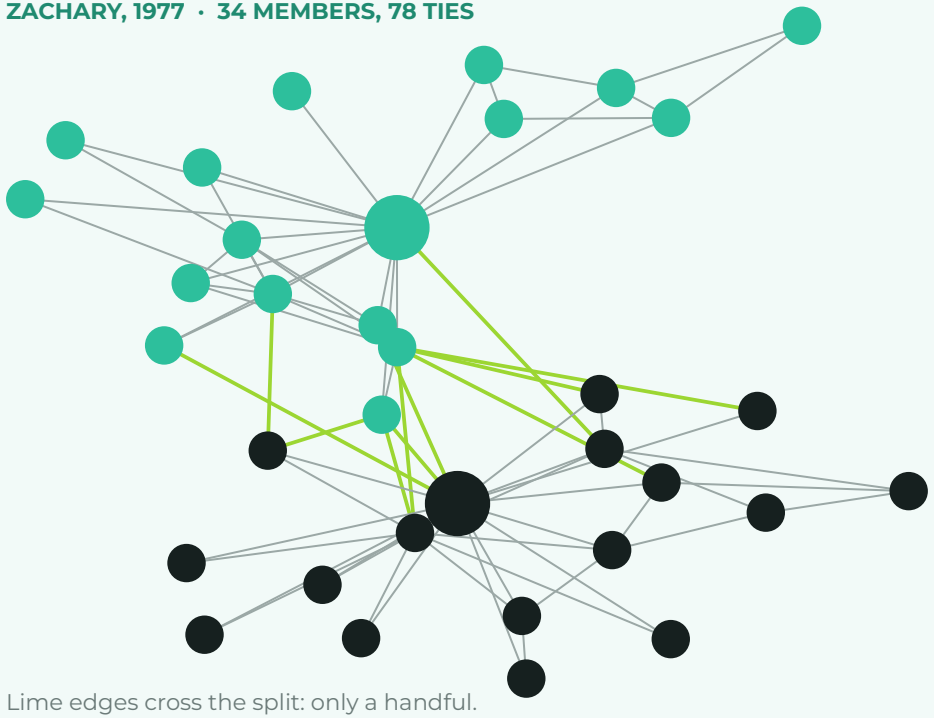
Communities are audiences that talk to each other. Platforms use them to recommend beyond your follows; analysts use them as segments; trust teams use odd ones to spot fake networks (later in this section).

SOURCE Girvan & Newman, PNAS 2002; Blondel et al., J. Stat. Mech. 2008; Traag, Waltman & van Eck, Sci. Reports 2019; Twitter Engineering (2023); Bluesky 2025 transparency report (Jan 2026).

COMMUNITIES · MODULARITY AND THE KARATE CLUB

The karate club split where the graph said it would.

ZACHARY, 1977 · 34 MEMBERS, 78 TIES



Lime edges cross the split: only a handful.

MODULARITY: HOW MUCH BETTER THAN RANDOM?

$$Q = (1/2m)\,\Sigma_{ij}\,[\,A_{ij} - k_i k_j / 2m\,]\cdot\delta(c_i, c_j)$$

Edges inside communities, minus the number you'd expect if edges were placed at random. Strong structure often scores 0.3–0.7.

- **The split:** a dispute between the instructor and the administrator divided the club (teal vs dark).
- **Girvan–Newman** recovered the split with only one member misclassified.
- **Q = 0.39** for the real split; Louvain finds four groups with Q = 0.44. Algorithms find structure, not necessarily the story.

SOURCE Zachary 1977; Girvan & Newman, PNAS 2002 (node 3 misclassified); Newman & Girvan, Phys. Rev. E 2004 (modularity); Q values calculated by the lecturer with NetworkX.

COMMUNITIES · CATCHING FAKE NETWORKS

Fake accounts give themselves away by shape and clicks.

GRAPH · SYBILRANK, NSDI 2012

Trust flows from real users

Fakes link to real people through few edges. Short random walks from known-real seeds give fakes little trust. At Tuenti, ~90% of the 200K accounts it ranked most suspicious were fake — versus ~5% from user reports.

CLICKS · USENIX SECURITY 2013

You are how you click

16,000 Renren users, 6.86M clicks: fakes browse differently. Clustering caught them with under 1% false positives; a scan of 1M users flagged 22K suspects; a LinkedIn test found 200 unknown fakes.

CAMPAIGNS · 2025–26

Coordinated behaviour

Accounts created together, posting in sync, linking mostly to each other. Platforms find them through behaviour, technical signals and network analysis; Bluesky removed 3,619 accounts for influence operations in 2025.

2026 twist: an LLM can write a perfect post. Timing, clicks and network shape are much harder to fake — remember Moltbook from Lecture 1.

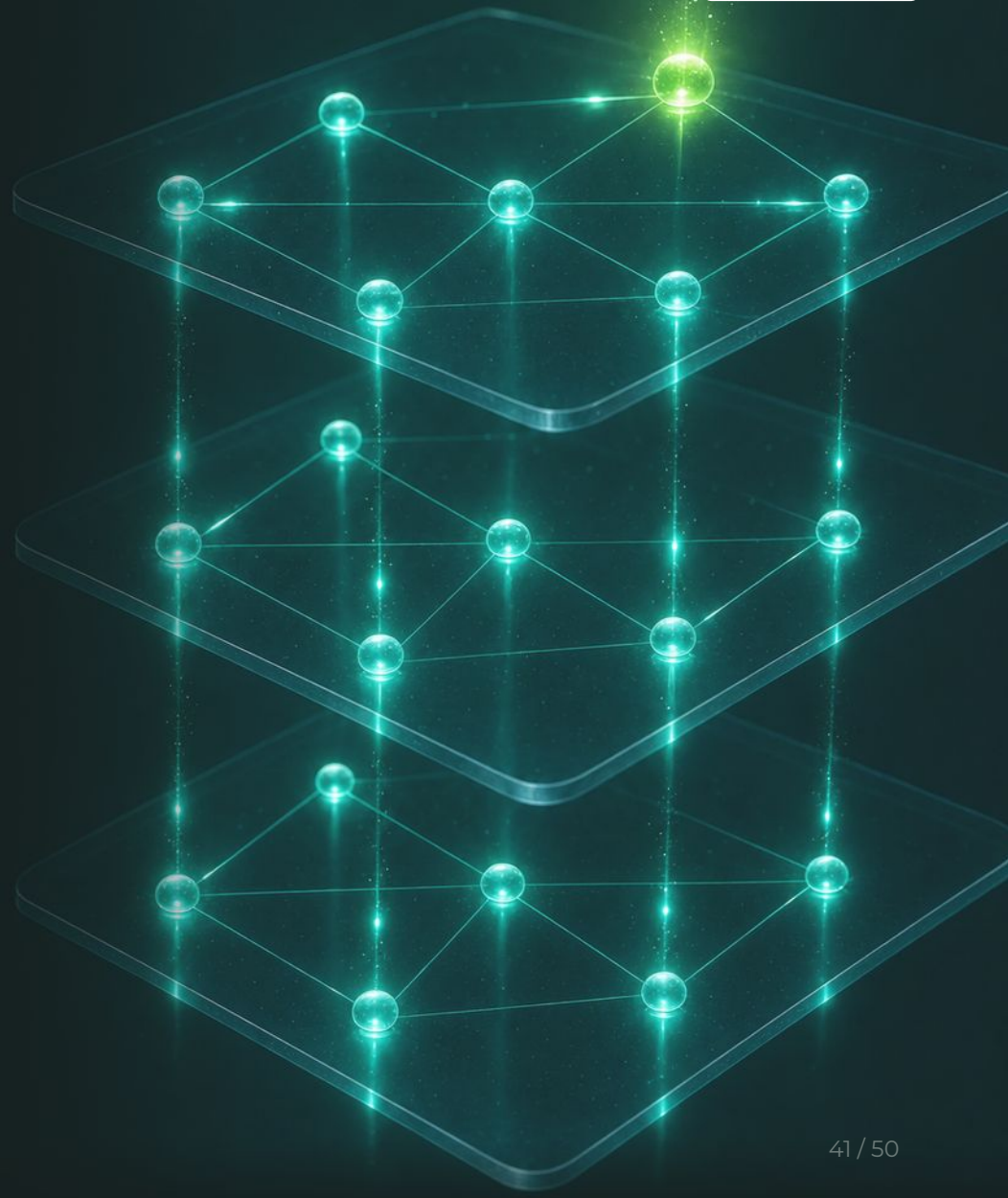
SOURCE Cao, Sirivianos, Yang & Pregueiro, NSDI 2012; Wang et al., “You are how you click”, USENIX Security 2013 (prototype tests); Meta adversarial threat reports; Bluesky 2025 transparency report.

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05

SECTION 5

Learning on graphs



PINTEREST · KDD 2018

PinSage trained a graph neural network on 3 billion nodes and 18 billion edges.

From hand-made graph features to embeddings and neural networks on graphs — the models behind recommendations, fraud detection and graph-powered retrieval.

NETWORKS · GRAPH MACHINE LEARNING

From graph statistics to graph neural networks.

2016

node2vec

Random walks turn the graph into “sentences” of nodes; word2vec turns them into node embeddings. Parameters p and q trade local for global structure.

2017+

Graph neural networks

Each node updates its vector from its neighbours' vectors, layer by layer (message passing). Two layers = friends of friends.

2018–26

GNNs in production

PinSage at Pinterest; LightGCN for collaborative filtering; graphs + LLMs for retrieval (GraphRAG). Details next.

Learn it properly: Stanford CS224W, Machine Learning with Graphs (J. Leskovec). This autumn's Lecture 11 — GNNs for recommender systems — ran yesterday, 27 October; the slides are on the course site.

SOURCE Grover & Leskovec, KDD 2016 (node2vec); Ying et al., KDD 2018 (PinSage); He et al., SIGIR 2020 (LightGCN); Stanford CS224W, Fall 2026 schedule.

GRAPH LEARNING · NODE EMBEDDINGS

Random walks turn a graph into sentences.

01

Walk

Start short random walks from every node.

02

Read

Each walk is a “sentence” of node IDs.

03

Embed

Skip-gram, as in word2vec: nodes that share walks get similar vectors.

04

Use

Link prediction, node classification, lookalike audiences.

ONE WALK = ONE SENTENCE



Same trick as Lecture 2's word2vec: “you shall know a node by the company it keeps.”

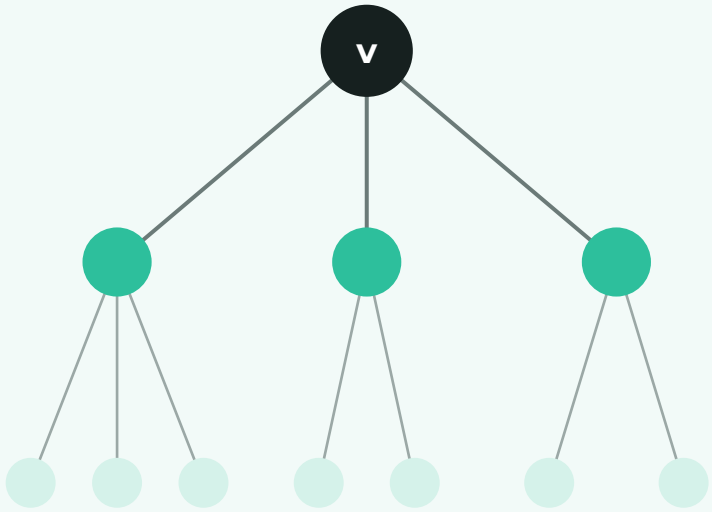
DEEPWALK 2014 → NODE2VEC 2016

DeepWalk uses uniform walks. node2vec adds two knobs, p and q: stay close to home (finds structural roles, like “broker”) or wander far (finds communities).

SOURCE Perozzi, Al-Rfou & Skiena, “DeepWalk”, KDD 2014; Grover & Leskovec, “node2vec”, KDD 2016; Stanford CS224W, Lecture 2.

GNNs: every node listens to its neighbours.

COMPUTATION TREE OF NODE v (2 LAYERS)



Layer 2 hears from friends of friends.

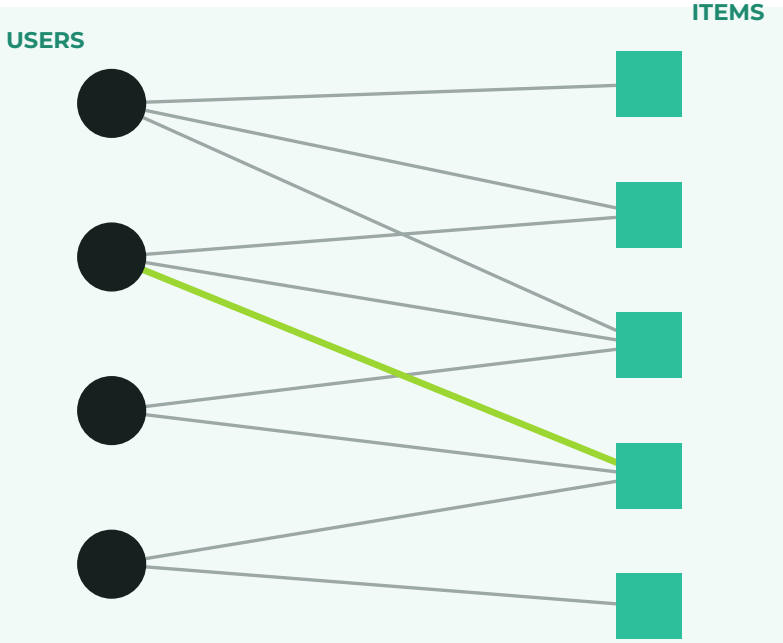
$$h_v^{(k+1)} = \sigma(W_\square \cdot \text{mean}_{u \in N(v)} h_u^{(k)} + B_\square \cdot h_v^{(k)})$$

- **Message passing:** each layer averages neighbours' vectors and mixes in the node's own.
- **k layers = k hops.** Too many and every node looks alike (over-smoothing).
- **GCN (ICLR 2017):** normalised neighbour averaging for semi-supervised node labels.
- **GraphSAGE (NeurIPS 2017):** samples and aggregates neighbours, so it works on brand-new nodes.

SOURCE Kipf & Welling, ICLR 2017 (GCN); Hamilton, Ying & Leskovec, NeurIPS 2017 (GraphSAGE); Stanford CS224W, Lectures 3–5.

GRAPH LEARNING · RECOMMENDATION

Recommendation is link prediction on a user–item graph.



Lime edge: the link to predict. Two hops away = a candidate.

PINSAGE · PINTEREST, KDD 2018

A GNN on 3 billion nodes

2B pins and 1B boards, 18B edges. Offline hit-rate 67%, +150% MRR over the best baseline; repins up 10–30% in A/B tests.

LIGHTGCN · SIGIR 2020

Less is more

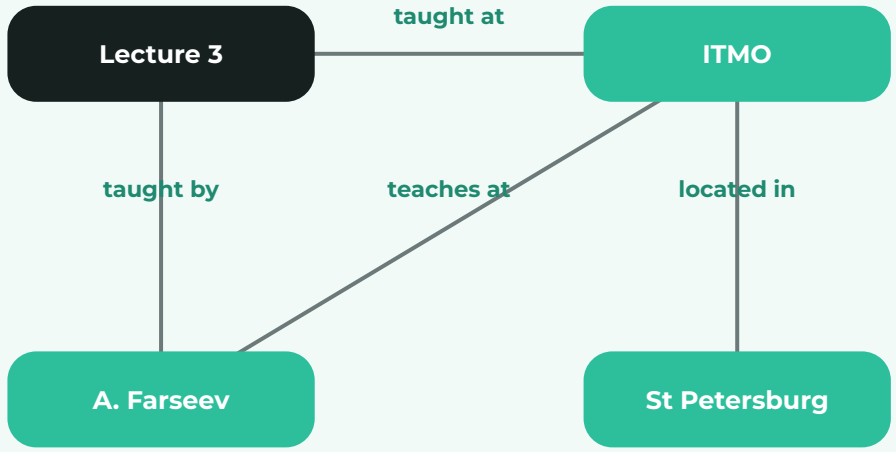
Strip the GNN down to neighbour averaging — no feature transforms, no non-linearities — and collaborative filtering gets better and simpler.

Why graphs help: they pass signal along paths — user → item → similar users → their items — so sparse rows borrow strength from the network. Section 4's random walks, trained end to end.

SOURCE Ying et al., "Graph Convolutional Neural Networks for Web-Scale Recommender Systems", KDD 2018; He et al., "LightGCN", SIGIR 2020; Stanford CS224W Lecture 11 (Fall 2025 slides).

Graphs give LLMs structure: knowledge graphs and GraphRAG.

A KNOWLEDGE GRAPH IS TRIPLES



Google's Knowledge Graph (2012): “things, not strings” — 500M+ objects, 3.5B+ facts at launch.

GRAPHRAG · MICROSOFT RESEARCH, 2024

01	**Extract**	An LLM reads the documents and builds an entity–relation graph.
02	**Cluster**	Leiden community detection (Section 4) groups related entities.
03	**Summarise**	The LLM writes a summary for each community.
04	**Answer**	Global questions combine the community summaries.

Part 1's RAG retrieves passages. GraphRAG retrieves structure.

SOURCE Singhal, “Introducing the Knowledge Graph: things, not strings”, Google blog (16 May 2012); Edge et al., “From Local to Global: A Graph RAG Approach” (Microsoft Research, 2024).

LECTURE 3, PART 2 · PUTTING IT TOGETHER

Five ideas to take from the graph.

01

Structure

Hubs and short paths are the norm, not the exception.

02

Ties

Strong ties close triangles; weak ties bridge worlds.

03

Influence

Topic and position beat follower counts.

04

Ranking

Whose links, not how many. Communities and fakes show in the shape.

05

Learning

Embeddings and GNNs turn the graph into features for search, feeds and ads.

Parts 1 and 2 together: every matching system reads text, behaviour and the graph. Last week was the engine; this week was the road network it drives on.

COURSE · FROM LECTURE TO PROJECT

Use the graph in Lab 2 and in your project.

LAB 2 · DUE SUN 6 DEC, 23:59

Multi-source prediction

Predicting virality? Add network features: the author's degree and core position, the speed of the first reshares, and the breadth of the early cascade.

MINI-PROJECT · TOPIC DUE SUN 8 NOV

One page per team

Problem, user, data, methods. Approved by Sun 15 Nov. Easiest open graph to start with: Bluesky's public firehose.

PROJECT IDEAS FROM TODAY

- ✓ Topic influencer finder on Bluesky (TwitterRank-style).
- ✓ Community map of a market's creators, as audience segments.
- ✓ Fake-account detector from timing and network shape.
- ✓ GraphRAG assistant over this course's slides.

SOURCE Course schedule (Lecture 1, slide 30); AT Protocol streaming docs (atproto.com). Deadlines are 23:59 Moscow time.

COURSE · GO DEEPER

Free reading from Stanford, MIT and the papers behind today.

01 GRAPH MACHINE LEARNING

Stanford CS224W (Fall 2026), J. Leskovec — node embeddings, GNNs, GNN recommenders (Lecture 11)

web.stanford.edu/class/cs224w

02 NETWORKS TEXTBOOK

Easley & Kleinberg — Networks, Crowds, and Markets (free): ch. 2–4, 14, 19–20

cs.cornell.edu/home/kleinber/networks-book

03 NETWORK SCIENCE

Barabási — Network Science (free online): ch. 2–5, graphs to scale-free networks

networksciencebook.com

04 NETWORKS COURSE

MIT 14.15J / 6.207J Networks (OCW, Spring 2018) — centrality, small worlds, diffusion

ocw.mit.edu/courses/14-15j-networks-spring-2018

05 INFLUENCE

Kempe, Kleinberg & Tardos — Maximizing the Spread of Influence through a Social Network (KDD 2003)

doi.org/10.1145/956750.956769

06 WEAK TIES, TESTED

Rajkumar et al. — A causal test of the strength of weak ties (Science, 2022)

doi.org/10.1126/science.abl4476



OTHER DETAILS · NEXT LESSON

Next week: Lecture 4 — evaluation metrics.

- **Mini-project topic:** one page per team by Sun 8 Nov, 23:59; approval by Sun 15 Nov.
- **Lab 2:** individual, due Sun 6 Dec, 23:59.
- **Office hours:** Prof. Aleks Farseev, 12 PM on Zoom, via Daniel Onishchenko or Jack Zhao.
- **Labs & project:** questions go to the tutors, Daniel Onishchenko and Jack Zhao.
- **Slides & reading:** posted on the course Telegram channel after class.

Next week: how to measure all of this — precision, nDCG, A/B tests, and what accuracy misses.

BEFORE LECTURE 4

Switch the algorithm off for a day.

Use the Following or non-personalised feed on one platform you use daily. Note what disappears, what you miss, and how long you stay. Bring one number: minutes on the app, before and after.